

Student Number _____

ASCHAM SCHOOL

**2021****YEAR 12****TRIAL****EXAMINATION**

Mathematics

Extension 2

General Instructions

- Reading time – 10 minutes.
- Working time – 3 hours.
- Write using black non-erasable pen.
- NESA-approved calculators may be used.
- A NESA Reference Sheet is provided.
- All necessary working should be shown in every question.

Total marks – 100

- Attempt Sections A and B.
- Section A is worth 10 marks.
- Recommended time on Section A: 15 minutes
- Answer Section A on the multiple choice answer sheet.
- Detach the multiple choice answer sheet from the back of the examination paper.
- Section B contains 6 questions worth 15 marks each.
- Recommended time on Section B: 2 hours 45 minutes
- Answer each question in a new booklet.
- Label all sections clearly with your name/number and teacher.

SECTION A – 10 MULTIPLE CHOICE QUESTIONS 10 MARKS**ANSWER ON THE ANSWER SHEET****1**

Which of the following is closest to the angle between the vector $\underline{u} = \begin{pmatrix} 3 \\ 5 \\ -2 \end{pmatrix}$ and the y -axis?

- A** 24°
- B** 36°
- C** 61°
- D** 109°

2

Which of the following is equivalent to the value of $\frac{2i}{e^{\frac{5\pi}{6}i}}$?

- A** $2\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$
- B** $2\left(\cos\left(-\frac{2\pi}{3}\right) + i\sin\left(-\frac{2\pi}{3}\right)\right)$
- C** $-2\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$
- D** $2\left(\cos\left(-\frac{2\pi}{3}\right) - i\sin\left(-\frac{2\pi}{3}\right)\right)$

3

Which of the following is true?

- A** $\forall x, y \in \mathbb{N}, \exists z \in \mathbb{Q} : z\sqrt{y} = \frac{x}{\sqrt{y}}$
- B** $\forall x, y \in \mathbb{R}, \exists z \in \mathbb{N} : z\sqrt{y} = \frac{x}{\sqrt{y}}$
- C** $\forall x, y \in \mathbb{Q}, \exists z \in \mathbb{Z} : z\sqrt{y} = \frac{x}{\sqrt{y}}$
- D** $\forall x, y \in \mathbb{Z}, \exists z \in \mathbb{R} : z\sqrt{y} = \frac{x}{\sqrt{y}}$

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4 Which of the following is a root of $z^5 + 1 = 0$?

A $\cos\left(\frac{\pi}{5}\right) + i \sin\left(\frac{\pi}{5}\right)$

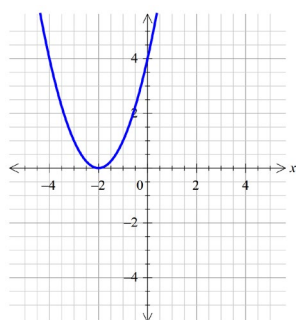
B $\cos\left(\frac{2\pi}{5}\right) + i \sin\left(\frac{2\pi}{5}\right)$

C $\cos\left(-\frac{2\pi}{5}\right) + i \sin\left(-\frac{2\pi}{5}\right)$

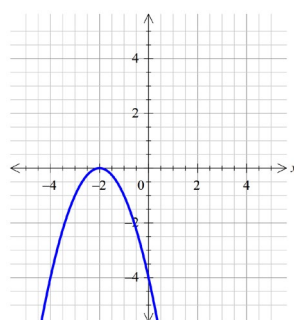
D $\cos\left(-\frac{4\pi}{5}\right) + i \sin\left(-\frac{4\pi}{5}\right)$

5 What is a possible graph of v^2 versus x for a particle moving in simple harmonic motion?

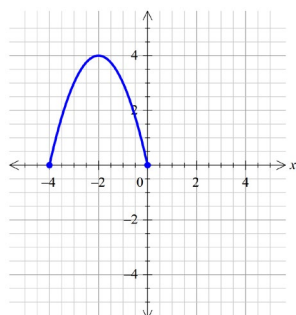
A



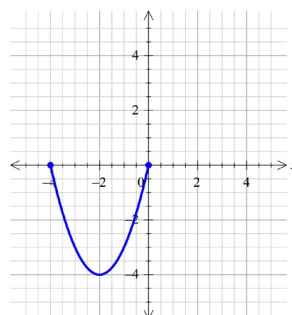
B



C



D

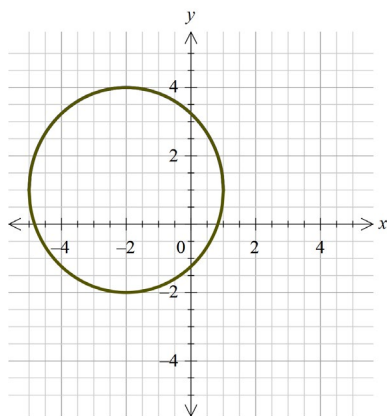


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- 6 Which of the following is a counter-example to the following statement?
If you are over 60 then you must get the AstraZeneca vaccine.

- A If you are under 60 then you must not get the AstraZeneca vaccine.
- B Jo is over 60 and got the Pfizer vaccine instead.
- C Polly is under 60 and got the AstraZeneca vaccine.
- D If you get the AstraZeneca vaccine you must be over 60.

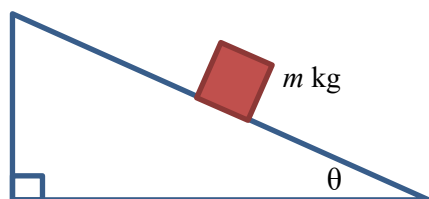
- 7 The graph below is on the Argand diagram.



What is the equation of the graph?

- A $|z - 2 - i| = 3$
- B $|z - 2 + i| = 9$
- C $|z + 2 - i| = 3$
- D $|z + 2 + i| = 9$

- 8 A mass m kg is resting on a slope inclined at θ to the horizontal. Forces acting are the normal force N , a frictional force F and the gravitational force mg .



Which of the following is the correct resolution of forces in the horizontal direction?

- A $F \cos \theta - N \sin \theta = 0$
 B $F \sin \theta - N \sin \theta = 0$
 C $F \sin \theta - N \cos \theta = 0$
 D $F \cos \theta - N \cos \theta = 0$
- 9 A particle P is moving in simple harmonic motion. Its maximum speed is 6π m/s and its displacement then is 4 m. After 10 seconds it has reached its maximum speed again. What is a possible equation for the displacement of P ?

- A $x = 60 \sin\left(\frac{\pi t}{10}\right) + 4$
 B $x = 60 \sin\left(\frac{\pi t}{20}\right) + 4$
 C $x = 60 \sin(10t) + 4$
 D $x = 60 \sin(20t) + 4$

- 10 Consider the complex numbers $z_1 = 1 - i$ and $z_2 = 1 + i\sqrt{3}$. What is the smallest positive value of n such that $\frac{(z_2)^n}{(z_1)^n}$ is purely imaginary?

- A 2
 B 4
 C 6
 D 8

SECTION 2 – 6 QUESTIONS EACH WORTH 15 MARKS**Question 11 – Begin a new writing booklet**

- a** Find $\int \frac{x+1}{x^2+4} dx$. **3**
- b** Find $\int_1^2 \frac{1}{x(x+5)} dx$. **4**
- c** Find $\int \tan^6 x \, dx$. **4**
- d** Calculate the modulus and argument of the product of the roots of the equation $(5+3i)z^2 - (1-4i)z + (8-2i) = 0$. **4**

Question 12 – Begin a new writing booklet

- a** $p \in \mathbb{N}$
Consider the statement for p :
If p has an odd number of distinct factors then p is a square number.
- i** Write the converse. **1**
- ii** Write the contrapositive. **2**
- iii** Write the negation. **2**
- iv** Determine whether or not the statement is an equivalence. Give reasons. Is the statement true? **2**
- b** Let $z = x + iy$ where $x, y \in \mathbb{R}$. Sketch on an Argand diagram the points satisfying $z^2 - \bar{z}^2 = 8i$. **2**
- c** Find all complex numbers $z = x + iy$ where $x, y \in \mathbb{R}$ such that $|z|^2 + 5\bar{z} + 10i = 0$. **3**
- d** Let $z_1, z_2 \in \mathbb{C}$ such that $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$. Prove that $|z_1||z_2| = |z_1 z_2|$. **3**

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Question 13 – Begin a new writing booklet**a**

The 4 complex numbers z_1, z_2, z_3, z_4 are represented by the points Z_1, Z_2, Z_3, Z_4 on an Argand diagram. If $z_1 - z_2 + z_3 - z_4 = 0$ and $z_1 - iz_2 - z_3 + iz_4 = 0$, what type of quadrilateral could $Z_1Z_2Z_3Z_4$ be? **3**

b

Consider the line l_1 with equation $\frac{x-2}{3} = \frac{y+4}{2} = \frac{z-1}{-4}$.

i

Find the vector equation of any line l_2 that passes through the point $P(1, 2, 3)$, which is perpendicular to l_1 . Is this solution unique? **3**

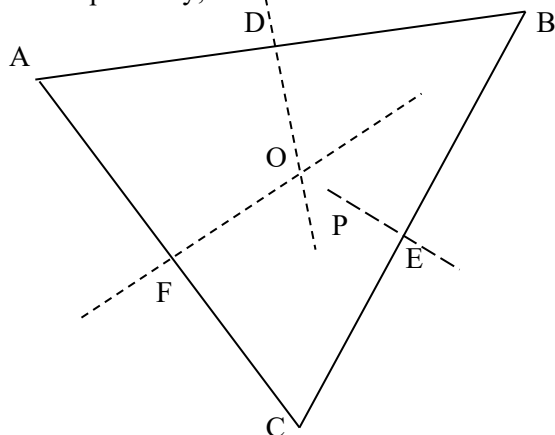
ii

If the lines l_1 and l_2 intersect, find the point of intersection. **3**

Question 13 continues on the next page.

Question 13 continued

- c** Consider the triangle ABC shown. The perpendicular bisectors, DO and FO , of AB and AC respectively, intersect at O .



Copy the diagram.

Let $\overrightarrow{OA} = \underline{a}$, $\overrightarrow{OB} = \underline{b}$, $\overrightarrow{OC} = \underline{c}$, $\overrightarrow{OD} = \underline{d}$, etc.

Use vectors to:

- | | | |
|------------|--|----------|
| i | prove that $\underline{d} = \frac{1}{2}(\underline{a} + \underline{b})$ | 1 |
| ii | prove that A, B and C all lie on a circle with centre O , ie $ \overrightarrow{OA} = \overrightarrow{OB} = \overrightarrow{OC} $ | 2 |
| iii | prove that the perpendicular bisector of BC , that is, EP passes through O . | 3 |

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Question 14 – Begin a new writing booklet

- a i** If a, b are real, show that $\frac{a^2 + b^2}{2} \geq ab$. **1**

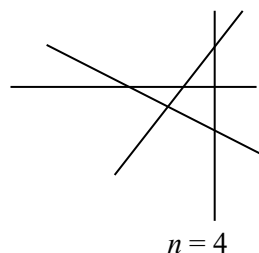
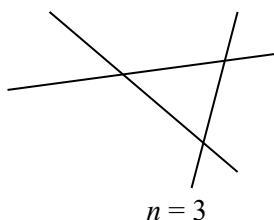
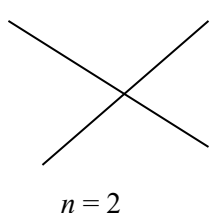
- ii** Hence, or otherwise, show that if a, b, c, d are real and $a, b, c, d \geq 0$ then **2**
 $\sqrt{(a^2 + c^2)(b^2 + d^2)} \geq ab + cd$.

- b** Prove that $\int_0^a f(x) dx = \int_0^a f(a-x) dx$ using the substitution $u = a-x$. **1**

Hence evaluate $\int_0^2 x(2-x)^{10} dx$. **3**

- c** Prove by the contrapositive that if m and n are positive integers such that $m+n$ and $m-n$ have no common factors, then m and n have no common factors. **3**

- d** Consider the maximum number N of possible points of intersection of n lines on a plane, as shown in the series of diagrams below.



- i** Copy and complete the table: **2**

n	2	3	4	5	6
N	1	3			

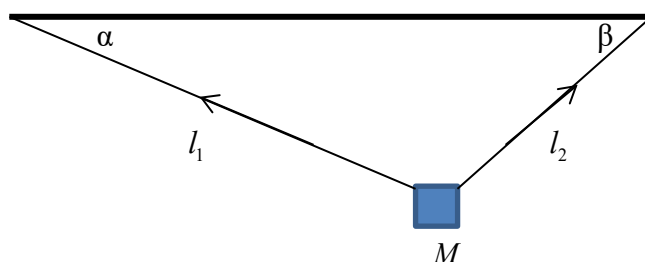
- ii** Prove by induction that the maximum number N of points of intersection of n lines is **3**

$$N = \frac{n(n-1)}{2} \text{ for } n \geq 2.$$

Question 15 – Begin a new writing booklet

- a** Find the four 4th roots of $-\frac{1}{2} + \frac{i\sqrt{3}}{2}$ and show them on an Argand diagram. **4**

- b** A body of mass M kg is being suspended by two strings of differing lengths l_1 and l_2 attached to the mass and then to either end of a horizontal rod as shown in the diagram. Acceleration due to gravity is g m/s². **4**



The tensions in l_1 and l_2 are T_1 and T_2 (Newtons) respectively.

Show that the tension T_1 in the string l_1 is given by

$$T_1 = \frac{Mg \cos \beta}{\sin(\alpha + \beta)}.$$

- c** Let $I_n = \int \frac{dx}{(x^2 + 1)^n}$.

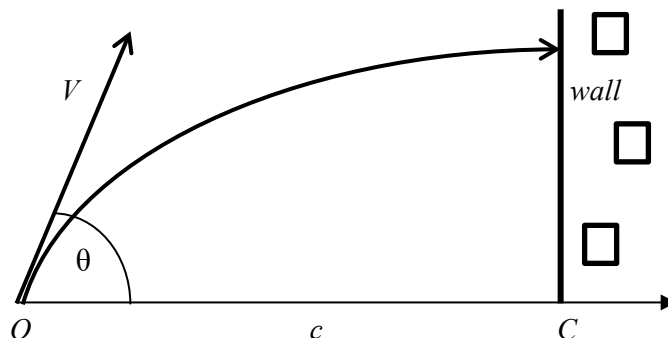
- i** Show that $\int \frac{dx}{(x^2 + 1)^n} = \int \frac{dx}{(x^2 + 1)^{n-1}} - \int \frac{x^2 dx}{(x^2 + 1)^n}$. **1**

- ii** Use the identity above to find constants A and B such that $I_n = \frac{Ax}{(x^2 + 1)^{n-1}} + BI_{n-1}$. **4**

- iii** Hence, or otherwise find $\lim_{k \rightarrow \infty} \int_0^k \frac{dx}{(x^2 + 1)^2}$. **2**

Question 16 – Begin a new writing booklet

- a** A hose is at O and aimed to hit a vertical wall which is c metres along the horizontal ground from O . The hose water comes out at a speed V m/s and is aimed at an angle θ from the horizontal where $0^\circ < \theta < 90^\circ$. Take g m/s² as acceleration due to gravity and ignore air resistance. **4**



Show that the hose water will reach the wall if and only if $V > \sqrt{gc}$.

- b i** Show that $1 + r + r^2 + r^3 + \dots + r^{n-1} = \frac{1-r^n}{1-r}$ for $r \neq 1$. **1**
- ii** As $n \rightarrow \infty$, find an expression for $1 + r + r^2 + r^3 + \dots + r^{n-1}$ for $0 < r < 1$. **1**
- iii** Prove that for $0 < x < 1$, $x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \dots + \frac{x^n}{n} + \dots = -\ln(1-x)$. **2**

Question 16 continues on the next page.

Question 16 continued

- c** Two particles P_1 and P_2 are moving with velocities v_1 and v_2 respectively in Simple Harmonic Motion along the same axis according to the equations

$$\begin{aligned} x_1 &= A + \sin nt \\ \ddot{x}_2 &= -n^2 x_2 \end{aligned} \quad \text{where } n, A \in \mathbb{R} \text{ and } n, A > 0, t \geq 0.$$

Initially, $x_1 = A$, $x_2 = 1$ and $v_2 = 0$.

Assume the value of A is such that the particles never meet.

- i** Show that $\dot{x}_1^2 = n^2 (1 - (x_1 - A)^2)$. **2**
- ii** Show that $\dot{x}_2^2 = n^2 (1 - x_2^2)$. **2**
- iii** Find the minimum distance between the two particles. **3**

The end! 😊

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Student Number

ASCHAM SCHOOL**YEAR 12 Trial Mathematics Extension 2 Exam****MULTIPLE-CHOICE ANSWER SHEET**

- | | | | | |
|-----|-------------------------|-------------------------|-------------------------|-------------------------|
| 1. | A ☐ | B ☐ | C ☐ | D ☐ |
| 2. | A ☐ | B ☐ | C ☐ | D ☐ |
| 3. | A ☐ | B ☐ | C ☐ | D ☐ |
| 4. | A ☐ | B ☐ | C ☐ | D ☐ |
| 5. | A ☐ | B ☐ | C ☐ | D ☐ |
| 6. | A ☐ | B ☐ | C ☐ | D ☐ |
| 7. | A ☐ | B ☐ | C ☐ | D ☐ |
| 8. | A ☐ | B ☐ | C ☐ | D ☐ |
| 9. | A ☐ | B ☐ | C ☐ | D ☐ |
| 10. | A ☐ | B ☐ | C ☐ | D ☐ |



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MULTIPLE CHOICE

1. y-axis is $(0, 1, 0)$ say.

$$\underline{u} \cdot \underline{y} = |\underline{u}| |\underline{y}| \cos \theta$$

$$\begin{aligned} \therefore \cos \theta &= \frac{3(0) + 5(1) - 2(0)}{\sqrt{3^2 + 5^2 + (-2)^2} \sqrt{0^2 + 1^2 + 0^2}} \\ &= \frac{5}{\sqrt{38}} \quad (B) \\ \theta &\approx 36^\circ \end{aligned}$$

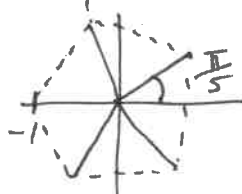
$$\begin{aligned} 2. \frac{2i}{e^{\frac{5\pi}{6}i}} &= \frac{2e^{\frac{\pi}{2}i}}{e^{\frac{5\pi}{6}i}} \\ &= 2e^{i(\frac{\pi}{2} - \frac{5\pi}{6})} \\ &= 2e^{i(-\frac{\pi}{3})} \\ &= 2\left(\cos\left(-\frac{\pi}{3}\right) + i\sin\left(-\frac{\pi}{3}\right)\right) \\ &= 2\left(\cos\frac{\pi}{3} - i\sin\left(\frac{\pi}{3}\right)\right) \\ &= 2\left(\frac{1}{2} - i\frac{\sqrt{3}}{2}\right) \\ &= 2\left(-\cos\frac{2\pi}{3} - i\sin\left(\frac{2\pi}{3}\right)\right) \\ &= (C) \end{aligned}$$

$$3. z\sqrt{y} = \frac{x}{\sqrt{y}} \text{ or } zy = x \text{ if } y > 0$$

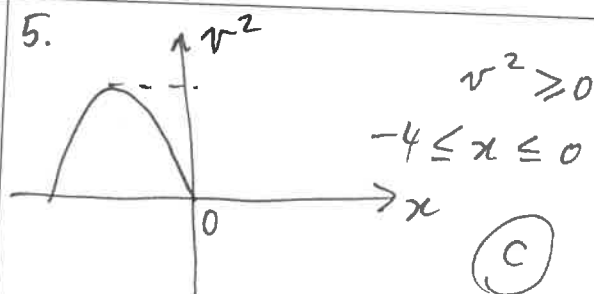
$$\text{or } z = \frac{x}{y} \\ \forall x, y \in \mathbb{N}, \exists z \in \mathbb{Q} : z\sqrt{y} = \frac{x}{\sqrt{y}}$$

$$4. z^5 + 1 = 0. \quad z^5 = -1$$

(A)

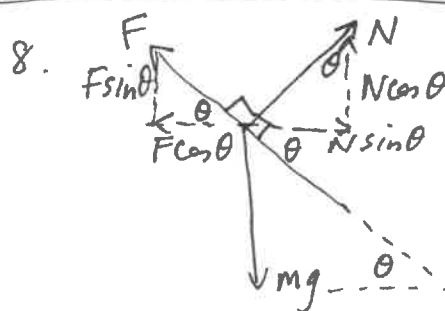


5.



6. (B) Jo is over 60 and did not get the Astra Z vaccine.

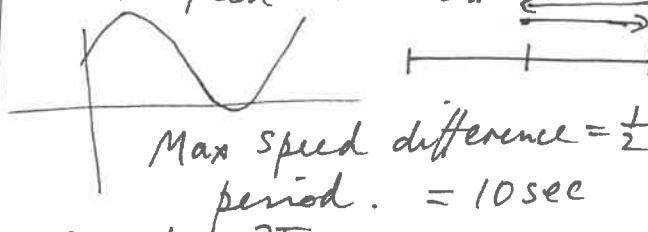
$$\begin{aligned} 7. \text{Centre } (-2, 1) \quad -2 + i, r = 3 \\ |z - (-2 + i)| = 3 \\ |z + 2 - i| = 3 \quad (C) \end{aligned}$$



$$\text{Horizontal: } F \cos \theta - N \sin \theta = 0$$

$$9. x = A \sin nt + k \\ \dot{x} = An \cos nt \quad -1 \leq \cos nt \leq 1$$

$$\text{Max speed} = An = 6\pi$$



$$\text{Period} = \frac{2\pi}{n}$$

$$20 = \frac{2\pi}{n} \therefore n = \frac{2\pi}{20} = \frac{\pi}{10}$$

$$\therefore A\left(\frac{\pi}{10}\right) = 6\pi \therefore A = 60.$$

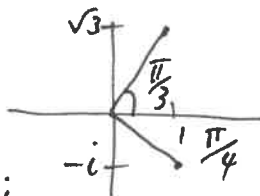
$$\therefore x = 60 \sin\left(\frac{\pi}{10} t\right) + 4 \quad (A)$$

MULT. CHOICE cont'd

10. $z_1 = 1-i$, $z_2 = 1+i\sqrt{3}$.

$$\frac{(z_2)^n}{(z_1)^n} = ki$$

$$\left(\frac{1+i\sqrt{3}}{1-i} \right)^n = ki \quad k \in \mathbb{R}$$



$$\left(\frac{2 \operatorname{cis} \frac{\pi}{3}}{\sqrt{2} \operatorname{cis} \frac{\pi}{4}} \right)^n = \left(\sqrt{2} \operatorname{cis} \left(\frac{\pi}{3} - \frac{\pi}{4} \right) \right)^n$$

$$= (\sqrt{2})^n \left(\operatorname{cis} \left(\frac{\pi}{12} \right) \right)^n$$

$$ki = (\sqrt{2})^n \left(\operatorname{cis} \frac{n\pi}{12} \right)$$

$$\operatorname{cis} \frac{\pi}{2} = \operatorname{cis} \frac{6\pi}{12} \text{ is imaginary}$$

$$\therefore n=6.$$

(C)

Q11

$$a) \int \frac{x+1}{x^2+4} dx = \int \frac{x}{x^2+4} + \frac{1}{x^2+4} dx$$

$$(3) = \frac{1}{2} \ln|x^2+4| + \frac{1}{2} \tan^{-1} \frac{x}{2} + C$$

$$b) \int_1^2 \frac{1}{x(x+5)} dx = \int_1^2 \frac{A}{x} + \frac{B}{x+5} dx$$

$$\text{So } 1 \equiv A(x+5) + Bx$$

$$x=-5 \quad 1 = Bx-5 \Rightarrow B = \frac{-1}{5}$$

$$x=0 \quad 1 = 5A \Rightarrow A = \frac{1}{5}$$

$$\therefore \int_1^2 \frac{1}{x(x+5)} dx = \int_1^2 \frac{1}{5} \frac{1}{x} - \frac{1}{5} \frac{1}{x+5} dx$$

$$= \frac{1}{5} \left[\ln|x| - \ln|x+5| \right]_1^2$$

$$(4) = \frac{1}{5} \left[\ln|2| - \ln|7| - (\ln|1| - \ln|6|) \right]$$

$$= \frac{1}{5} (\ln|2| - \ln|7| + \ln|6|)$$

$$= \frac{1}{5} \ln \left| \frac{12}{7} \right|$$

$$c) \int \tan^6 x dx = \int \tan^4 x \cdot \tan^2 x dx$$

$$= \int \tan^4 x (\sec^2 x - 1) dx \quad (4)$$

$$= \int \tan^4 x \sec^2 x - \int \tan^4 x dx$$

$$= \frac{\tan^5 x}{5} - \int \tan^2 x (\sec^2 x - 1) dx$$

$$= \frac{\tan^5 x}{5} - \int \tan^2 x \sec^2 x - \int \tan^2 x dx$$

$$= \frac{\tan^5 x}{5} - \frac{\tan^3 x}{3} - \int \sec^2 x - 1 dx$$

$$= \frac{\tan^5 x}{5} - \frac{\tan^3 x}{3} + \tan x - x + C$$

$$d) (5+3i)z^2 - (1-4i)z + (8-2i) = 0$$

$$\text{Product of roots} = \frac{c}{a}$$

$$= \frac{8-2i}{5+3i} \times \frac{5-3i}{5-3i}$$

$$= \frac{40-6-10i-24i}{25+9}$$

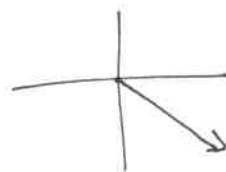
(4)

$$= \frac{34-34i}{34}$$

$$= 1-i$$

$$|1-i| = \sqrt{1+1} = \sqrt{2}$$

$$\arg(1-i) = -\frac{\pi}{4}$$



Q12

a) i) Converse of $P \Rightarrow Q$ is $Q \Rightarrow P$.
If p is a square number then p has an odd number of distinct factors.

ii) Contrapositive of $P \Rightarrow Q$ is
 $\neg Q \Rightarrow \neg P$. (1)

If p is not a square number then p does not have an odd number of distinct factors. (2)

iii) The negation: (2)

There exists ~~an~~ a number p with an odd number of distinct factors such that p is not a square number.

iv) It is an equivalence because both $P \Rightarrow Q$ AND $\neg Q \Rightarrow \neg P$ are true. The statement is true. (2)

eg. $36 = 6^2$. Factors come in pairs except for 6:
1, 36, 18, 2, 12, 3, 9, 4, 6.

b) $z^2 - \bar{z}^2 = 8i$.

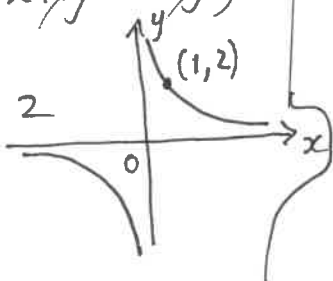
$(x+iy)^2 - (x-iy)^2 = 8i$ (2)

$(x+iy - x+iy)(x+iy + x-iy) = 8i$

$2iy \times 2x = 8i$

$4ixy = 8i$ 2

$xy = 2$.



1a) $|z|^2 + 5\bar{z} + 10i = 0$

$|x+iy|^2 + 5(x-iy) = -10i$

$x^2 + y^2 + 5x - 5iy + 10i = 0$

Equate reals and imaginaries:

$x^2 + y^2 + 5x = 0$, $-5iy + 10i = 0$

$x^2 + y^2 + 5x = 0$

$5y = 10$
 $y = 2$

$x^2 + 5x + 4 = 0$

$(x+4)(x+1) = 0$

$x = -4$ or $x = -1$, $y = 2$.

$\therefore z = -4 + 2i$, or $-1 + 2i$

d) RTP: $|z_1||z_2| = |z_1 z_2|$

Proof: LHS = $|z_1||z_2|$

$= |x_1 + iy_1||x_2 + iy_2|$

(3) $= \sqrt{x_1^2 + y_1^2} \sqrt{x_2^2 + y_2^2}$

$= \sqrt{(x_1^2 + y_1^2)(x_2^2 + y_2^2)}$

RHS = $|z_1 z_2|$

$= |(x_1 + iy_1)(x_2 + iy_2)|$

$= |x_1 x_2 + x_1 i y_2 + x_2 i y_1 + i^2 y_1 y_2|$

$= |(x_1 x_2 - y_1 y_2) + i(x_1 y_2 + x_2 y_1)|$

$= \sqrt{(x_1 x_2 - y_1 y_2)^2 + (x_1 y_2 + x_2 y_1)^2}$

$= \sqrt{x_1^2 x_2^2 - 2x_1 x_2 y_1 y_2 + y_1^2 y_2^2 + x_1^2 y_2^2 + 2x_1 x_2 y_1 y_2 + x_2^2 y_1^2}$

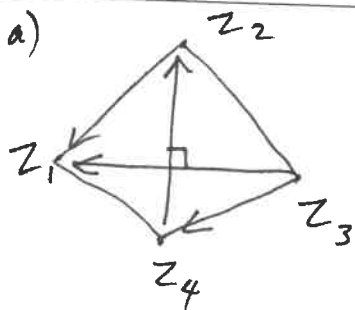
$= \sqrt{x_1^2 x_2^2 + x_1^2 y_2^2 + y_1^2 y_2^2 + y_1^2 x_2^2}$

$= \sqrt{x_1^2(x_2^2 + y_2^2) + y_1^2(y_2^2 + x_2^2)}$

$= \sqrt{(x_2^2 + y_2^2)(x_1^2 + y_1^2)}$

$= \text{LHS}$
 $\therefore |z_1||z_2| = |z_1 z_2|$

13 a)



34

$$z_1 - z_2 + z_3 - z_4 = 0$$

$$\therefore z_1 - z_2 = z_4 - z_3$$

$\therefore z_1 - z_2$ is parallel to $z_4 - z_3$

$$\text{and } |z_1 - z_2| = |z_4 - z_3|$$

$\therefore z_1, z_2, z_3, z_4$ has one pair of opposite sides equal and parallel $\therefore$ could be a parm.

$$\text{Also, } z_1 - iz_2 - z_3 + iz_4 = 0$$

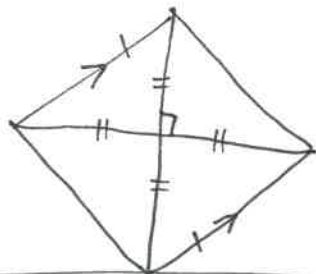
$$\therefore z_1 - z_3 = i(z_2 - z_4)$$

$\therefore z_1 - z_3$ is same as $z_2 - z_4$

only perpendicular $\therefore$

z_1, z_2, z_3, z_4 has equal diagonals that are perpendicular as well as a pair of opposite sides equal and parallel

It must be a square.



b) $l_1: \frac{x-2}{3} = \frac{y+4}{2} = \frac{z-1}{-4} = \lambda$

i) $l_1: x = 3\lambda + 2$

$$y = 2\lambda - 4$$

$$z = -4\lambda + 1$$

$$\text{or } \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ -4 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 2 \\ -4 \end{pmatrix}$$

3

Let direction be $\perp$ to $l_1 \therefore$

l_2 direction is $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ where

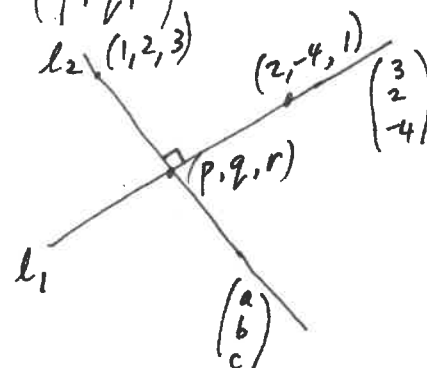
$$\begin{pmatrix} 3 \\ 2 \\ -4 \end{pmatrix} \cdot \begin{pmatrix} a \\ b \\ c \end{pmatrix} = 0 \text{ i.e. } 3a + 2b - 4c = 0$$

Say $a=2, b=3, c=3$

$$\text{then } l_2 \text{ is } \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda_1 \begin{pmatrix} 2 \\ 3 \\ 3 \end{pmatrix}$$

Not a unique solution

ii) l_1 and l_2 intersect at say (p, q, r) and $l_1 \perp l_2$ so



3

$$\exists a, b, c \text{ such that } \begin{matrix} a = p-1 \\ b = q-2 \\ c = r-3 \end{matrix}$$

$$\text{and } 3a + 2b - 4c = 0$$

$$\text{and } \begin{pmatrix} p \\ q \\ r \end{pmatrix} = \begin{pmatrix} 2 \\ -4 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 2 \\ -4 \end{pmatrix}$$

$$\therefore \begin{matrix} p = 2 + 3\lambda \\ q = -4 + 2\lambda \\ r = 1 - 4\lambda \end{matrix}$$

$$\text{so } \begin{matrix} a = 2 + 3\lambda - 1 = 1 + 3\lambda \\ b = -4 + 2\lambda - 2 = -6 + 2\lambda \\ c = 1 - 4\lambda - 3 = -2 - 4\lambda \end{matrix}$$

$$\therefore 3a + 2b - 4c = 0$$

$$3(1 + 3\lambda) + 2(-6 + 2\lambda) - 4(-2 - 4\lambda) = 0$$

$$3 + 9\lambda - 12 + 4\lambda + 8 + 16\lambda = 0$$

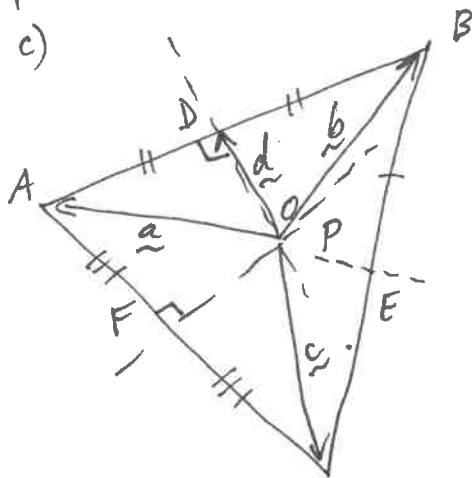
$$-1 + 29\lambda = 0$$

$$\therefore \lambda = \frac{1}{29}$$

$$\therefore \text{Point is } \left(2 + \frac{3}{29}, -4 + \frac{2}{29}, 1 - \frac{4}{29} \right) = \left(\frac{2}{29}, -3\frac{27}{29}, \frac{25}{29} \right)$$

Q13 cont'd

c)

i) RTP: $\underline{d} = \frac{1}{2}(\underline{a} + \underline{b})$ Proof: Given $\overrightarrow{AD} = \frac{1}{2} \overrightarrow{AB}$

$$\therefore \underline{d} - \underline{a} = \frac{1}{2}(\underline{b} - \underline{a})$$

$$\underline{d} = \frac{1}{2}\underline{b} - \frac{1}{2}\underline{a} + \underline{a} \quad (1)$$

$$= \frac{1}{2}\underline{b} + \frac{1}{2}\underline{a}$$

$$= \frac{1}{2}(\underline{a} + \underline{b}) \quad \text{QED!!!}$$

ii) RTP: $|\overrightarrow{OA}| = |\overrightarrow{OB}| = |\overrightarrow{OC}|$ Proof: Given $\overrightarrow{OD} \perp \overrightarrow{AB}$

$$\text{then } \overrightarrow{OD} \cdot \overrightarrow{AB} = 0$$

$$(\underline{d} - \underline{o}) \cdot (\underline{b} - \underline{a}) = 0$$

$$\therefore \frac{1}{2}(\underline{a} + \underline{b}) \cdot (\underline{b} - \underline{a}) = 0$$

$$\frac{1}{2}(\underline{b} \cdot \underline{b} - \underline{a} \cdot \underline{a}) = 0$$

$$\therefore |\underline{b}|^2 - |\underline{a}|^2 = 0$$

$$\therefore |\underline{b}|^2 = |\underline{a}|^2 \quad (|\underline{a}|, |\underline{b}| > 0)$$

$$\therefore |\underline{b}| = |\underline{a}|$$

$$\therefore |\overrightarrow{OB}| = |\overrightarrow{OA}|$$

Similarly we can see that

$$\underline{f} = \frac{1}{2}(\underline{a} + \underline{c})$$

Given $\overrightarrow{OF} \perp \overrightarrow{AC}$

$$\text{then } \overrightarrow{OF} \cdot \overrightarrow{AC} = 0$$

$$\therefore \underline{f} \cdot (\underline{c} - \underline{a}) = 0$$

$$\therefore \frac{1}{2}(\underline{a} + \underline{c}) \cdot (\underline{c} - \underline{a}) = 0$$

$$\therefore \underline{c} \cdot \underline{c} - \underline{a} \cdot \underline{a} = 0 \quad (2)$$

$$\therefore |\underline{c}|^2 = |\underline{a}|^2$$

$$\therefore |\overrightarrow{OC}| = |\overrightarrow{OA}| = |\overrightarrow{OB}| \quad \text{QED!}$$

iii) RTP: $\overrightarrow{PE} = \lambda \overrightarrow{EO}$ Proof: We know that $\underline{e} = \frac{1}{2}(\underline{c} + \underline{b})$

$$\text{Also } |\overrightarrow{OC}| = |\overrightarrow{OB}|$$

If $\overrightarrow{EP}$ is the perp. bisector then

$$\overrightarrow{EP} \cdot \overrightarrow{BC} = 0$$

$$\therefore (\underline{p} - \underline{e}) \cdot (\underline{c} - \underline{b}) = 0$$

$$\therefore (\underline{p} - \frac{1}{2}(\underline{c} + \underline{b})) \cdot (\underline{c} - \underline{b}) = 0$$

$$\therefore \underline{p} \cdot (\underline{c} - \underline{b}) - \frac{1}{2}(\underline{c} \cdot \underline{c} - \underline{b} \cdot \underline{b}) = 0$$

$$\therefore \underline{p} \cdot (\underline{c} - \underline{b}) - \frac{1}{2}(|\underline{c}|^2 - |\underline{b}|^2) = 0$$

$$\text{but } |\underline{c}|^2 = |\underline{b}|^2$$

$$\therefore \underline{p} \cdot (\underline{c} - \underline{b}) - \frac{1}{2}(0) = 0 \quad (3)$$

$$\therefore \underline{p} \cdot (\underline{c} - \underline{b}) = 0$$

$$\therefore (\underline{p} - \underline{o}) \cdot (\underline{c} - \underline{b}) = 0$$

$$\therefore \overrightarrow{OP} \perp \overrightarrow{CB}$$

but $\overrightarrow{EP} \perp \overrightarrow{CB}$ as well

$$\therefore P \text{ must lie on } \overrightarrow{OE}$$

$$\text{or } \overrightarrow{PE} = \lambda \overrightarrow{EO}$$

$$\therefore EP \text{ passes through } O.$$

Q14

a i) RTP: $\frac{a^2+b^2}{2} \geq ab$

Proof: Consider the difference:

$$\frac{a^2+b^2}{2} - ab = \frac{a^2+b^2-2ab}{2} = \frac{(a-b)^2}{2} \quad (1)$$

≥ 0 since $(a-b)^2 \geq 0$

$$\therefore \frac{a^2+b^2}{2} \geq ab.$$

ii) RTP: $\sqrt{(a^2+c^2)(b^2+d^2)} \geq ab+cd.$

Proof: Using the fact that if $\sqrt{P} \geq Q$ and $P, Q \geq 0$

then $P \geq Q^2$

and if $P \geq Q^2$ and $P, Q \geq 0$

then $\sqrt{P} \geq Q.$

So RTP: $(a^2+c^2)(b^2+d^2) \geq (ab+cd)^2$

Proof: consider the difference.

$$(a^2+c^2)(b^2+d^2) - (ab+cd)^2$$

$$= a^2b^2 + a^2d^2 + c^2b^2 + c^2d^2 - (2abcd + a^2b^2 + c^2d^2)$$

$$= \cancel{a^2b^2} + \cancel{a^2d^2} + \cancel{c^2b^2} + \cancel{c^2d^2} - 2abcd$$

$$= a^2d^2 + c^2b^2 - 2abcd \quad (2)$$

$$= (ad+cb)^2$$

≥ 0 since square

$$\therefore (a^2+c^2)(b^2+d^2) \geq (ab+cd)^2$$

$$\therefore \sqrt{(a^2+c^2)(b^2+d^2)} \geq ab+cd.$$

(since $a, b, c, d \geq 0$)

14

b) i) RTP: $\int_0^a f(x) dx = \int_0^a f(a-x) dx$

Let $u = a-x$ then $x=0 \Rightarrow u=a$
 $du = -1dx$ $x=a \Rightarrow u=0$

$$\therefore \text{RHS} = \int_0^a f(a-x) dx$$

$$= \int_a^0 f(u) \cdot -1 du \quad (1)$$

$$= \int_0^a f(u) du$$

$$= \int_0^a f(x) dx \quad (\text{dummy variable})$$

b) ii) $\int_0^2 x(2-x)^{10} dx$ let $a=2$

$$= \int_0^2 (2-x)(2-(2-x))^{10} dx$$

$$= \int_0^2 (2-x)(x)^{10} dx \quad (3)$$

$$= \int_0^2 2x^{10} - x^{11} dx$$

$$= \left[\frac{2x^{11}}{11} - \frac{x^{12}}{12} \right]_0^2$$

$$= \frac{2(2^{11})}{11} - \frac{2^{12}}{12} - (0-0)$$

$$= \frac{12(2^{12}) - 11(2^{12})}{132}$$

$$= \frac{2^{12}}{132}$$

$$= \frac{2^{10}}{33}$$

14 cont'd

c) Proof by Contrapositive:

RTP: If m, n are integers with ~~a~~ common factors then $m+n$ and $m-n$ also have a common factor.

Proof: Let $m = kp$ and

$n = kq$ where p, q are integers and k is an integer and the common factor of m and n ,

$$\text{then } m+n = kp+kq = k(p+q) \quad (3)$$

$$\text{and } m-n = kp-kq = k(p-q)$$

$\therefore m+n$ and $m-n$ both have common factor k .

$\therefore$ The contrapositive is true.

$\therefore$ Original statement: if $m+n$ and $m-n$ have no common factors then m and n have none. QED!

d) i)

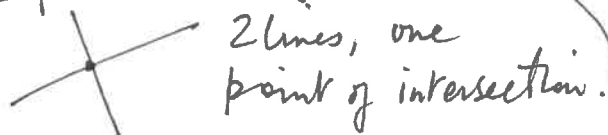
n	2	3	4	5	6
N	1	3	6	10	15

2 3 4 5

(2)

ii) RTP: $N = \frac{n(n-1)}{2}, n \geq 2$

Proof: Prove $P(2)$ true.



$$\begin{aligned} \therefore N &= \frac{2(2-1)}{2} \\ &= \frac{2 \times 1}{2} \\ &= 1 \end{aligned}$$

$\therefore P(2)$ true.

Assume $P(k)$ is true for some $k \in \mathbb{N}$,

i.e. $N = \frac{k(k-1)}{2}$ i.e. if there are k lines then $\frac{k(k-1)}{2}$ points of intersection

$$N = \frac{k(k-1)}{2}$$

RTP: $P(k+1)$ is true i.e. if $k+1$ lines then $N = \frac{(k+1)(k+1-1)}{2} = \frac{(k+1)k}{2}$ points

of intersection.

Proof: Consider adding another line to k lines. This creates another k points of intersection.

$$\begin{aligned} \therefore N &= \frac{k(k-1)}{2} + k \\ &= \frac{k(k-1)}{2} + \frac{2k}{2} \\ &= \frac{k^2 - k + 2k}{2} \\ &= \frac{k^2 + k}{2} \\ &= \frac{k(k+1)}{2} \end{aligned} \quad (3)$$

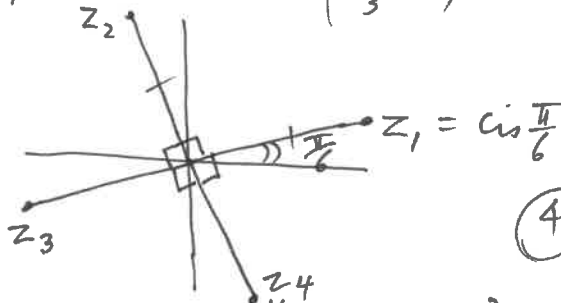
= RHS of $P(k+1)$.

$\therefore P(k+1)$ is true.

$\therefore P(n)$ true $\forall n \in \mathbb{N}, n \geq 2$ by Math Induction.

Q15

a) Let $z^4 = -\frac{1}{2} + \frac{i\sqrt{3}}{2} = 1 \operatorname{cis} \frac{2\pi}{3}$
 $z_1 = \text{1st root} = \operatorname{cis} \left(\frac{2\pi}{3} \div 4 \right) = \operatorname{cis} \frac{\pi}{6}$

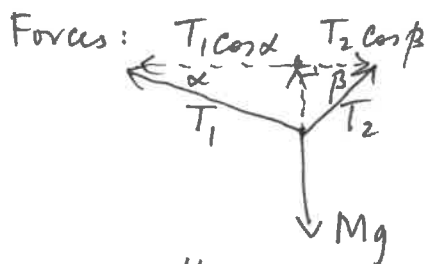
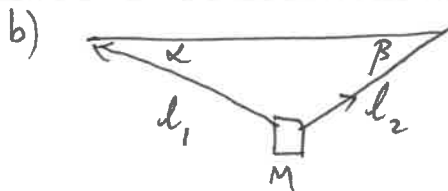


others equally spaced $\frac{2\pi}{4} = \frac{\pi}{2}$
 from z_1 .

$$z_2 = \operatorname{cis} \left(\frac{\pi}{6} + \frac{\pi}{2} \right) = \operatorname{cis} \frac{2\pi}{3}$$

$$z_3 = \operatorname{cis} \left(\frac{\pi}{6} + \pi \right) = \operatorname{cis} \frac{7\pi}{6}$$

$$z_4 = \operatorname{cis} \left(\frac{\pi}{6} - \frac{\pi}{2} \right) = \operatorname{cis} \left(-\frac{\pi}{3} \right)$$



Horizontally:

$$T_1 \cos \alpha - T_2 \cos \beta = 0$$

$$\therefore T_1 \cos \alpha = T_2 \cos \beta \Rightarrow T_2 = \frac{T_1 \cos \alpha}{\cos \beta}$$

Vertically:

$$T_1 \sin \alpha + T_2 \sin \beta - Mg = 0$$

$$\therefore T_1 \sin \alpha + \frac{T_1 \cos \alpha}{\cos \beta} \sin \beta = Mg$$

$$T_1 \sin \alpha \cos \beta + T_1 \cos \alpha \sin \beta = Mg \cos \beta$$

$$T_1 (\sin \alpha \cos \beta + \cos \alpha \sin \beta) = Mg \cos \beta$$

15 b) Cont'd:

$$T_1 (\sin(\alpha + \beta)) = Mg \cos \beta$$

$$\therefore T_1 = \frac{Mg \cos \beta}{\sin(\alpha + \beta)} \quad \text{QED!}$$

c) $I_n = \int \frac{dx}{(x^2+1)^n}$

i) RTP: $\int \frac{dx}{(x^2+1)^n} = \int \frac{dx}{(x^2+1)^{n-1}} - \int \frac{x^2 dx}{(x^2+1)^n}$

Proof: RHS = $\int \frac{dx}{(x^2+1)^{n-1}} - \int \frac{x^2 dx}{(x^2+1)^n}$

$$= \frac{\int (x^2+1) dx}{(x^2+1)^n} - \int \frac{x^2 dx}{(x^2+1)^n}$$

$$= \int \frac{x^2+1-x^2 dx}{(x^2+1)^n}$$

$$= \int \frac{dx}{(x^2+1)^n} = \text{LHS} \quad \text{QED!}$$

ii) $I_n = \frac{Ax}{(x^2+1)^{n-1}} + B I_{n-1}$

ie. $I_n = \frac{Ax}{(x^2+1)^{n-1}} + B \int \frac{dx}{(x^2+1)^{n-1}}$

Now: $I_n = \int \frac{dx}{(x^2+1)^{n-1}} - \int \frac{x^2 dx}{(x^2+1)^n}$

$$= \int \frac{dx}{(x^2+1)^{n-1}} - \int \frac{x \cdot x dx}{(x^2+1)^n}$$

$$= \int \frac{dx}{(x^2+1)^{n-1}} - \left[u x - \int v du \right]$$

where $u = x$
 $du = 1 dx$

$$dv = \frac{1}{2} \cdot \frac{2x dx}{(x^2+1)^n}$$

$$= \frac{1}{2} \cdot \frac{2x (x^2+1)^{-n} dx}{(x^2+1)^n}$$

P.T.O.

Q15 Cont'd

$$c) ii) \int \frac{dx}{(x^2+1)^n} = \int \frac{1 dx}{(x^2+1)^{n-1}} - \int \frac{x \cdot x dx}{(x^2+1)^n}$$

$$\text{Now } \int \frac{1 dx}{(x^2+1)^{n-1}} = I_{n-1}$$

$$\text{So } \int \frac{x \cdot x dx}{(x^2+1)^n} = \int u dv$$

$$= uv - \int v du \quad \text{Let } u = x$$

$$du = dx$$

$$dv = x(x^2+1)^{-n} dx$$

$$v = \frac{1}{2(-n+1)} (x^2+1)^{-n+1}$$

$$= \frac{1}{2(1-n)(x^2+1)^{n-1}}$$

$$= x \cdot \frac{1}{2(1-n)(x^2+1)^{n-1}} - \int \frac{1}{2(1-n)(x^2+1)^{n-1}} dx$$

$$= \frac{x}{2(1-n)(x^2+1)^{n-1}} + \frac{1}{2n-2} \int \frac{1 dx}{(x^2+1)^{n-1}}$$

$$\therefore I_n = I_{n-1} - \frac{x}{2(1-n)(x^2+1)^{n-1}} + \frac{1}{2n-2} I_{n-1}$$

$$= \frac{x}{(2n-2)(x^2+1)^{n-1}} + \frac{(2n-2)I_{n-1} + I_{n-1}}{2n-2}$$

$$= \frac{x}{(2n-2)(x^2+1)^{n-1}} + \frac{2n-3}{2n-2} I_{n-1}$$

$$\therefore A = \frac{1}{2n-2}, \quad B = \frac{2n-3}{2n-2}$$

Q15 cont'd

$$c) \text{ iii) } \lim_{k \rightarrow \infty} \int_0^k \frac{dx}{(x^2+1)^2} \quad n=2$$

$$= \lim_{k \rightarrow \infty} \left[\frac{1}{2(2)-2} \left[\frac{x}{(x^2+1)'} \right]_0^k + \frac{2(2)-3}{2(2)-2} \int_0^k \frac{1}{(x^2+1)'} dx \right]$$

$$= \lim_{k \rightarrow \infty} \left[\frac{1}{2} \left[\frac{k}{k^2+1} - 0 \right] + \frac{1}{2} \left[\tan^{-1} x \right]_0^k \right]$$

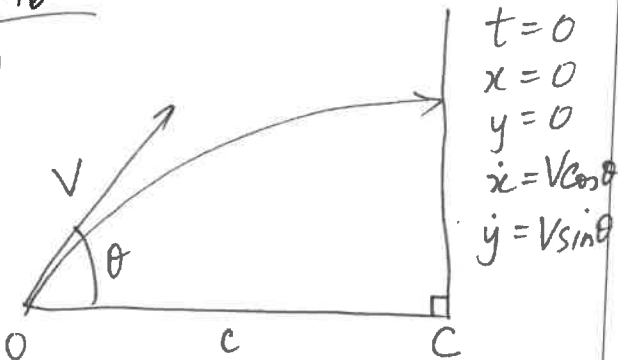
$$= \frac{1}{2}(0) + \frac{1}{2} \left(\frac{\pi}{2} - 0 \right)$$

$$= \frac{\pi}{4}$$

(2)

Q16

a)

Let $\ddot{x} = 0$, $\ddot{y} = -g$

$$\dot{x} = \int 0 dt, \quad \dot{y} = \int -g dt$$

$$= C_1, \quad = -gt + C_2$$

When $t=0$,

$$\dot{x} = V \cos \theta = C_1, \quad \dot{y} = V \sin \theta = -g(0) + C_2$$

$$\therefore \dot{x} = V \cos \theta, \quad \dot{y} = -gt + V \sin \theta$$

$$x = \int V \cos \theta dt, \quad y = \int -gt + V \sin \theta dt$$

$$x = Vt \cos \theta + C_3, \quad y = -\frac{gt^2}{2} + Vt \sin \theta + C_4$$

When $t=0$,

$$x = 0 = V(0) \cos \theta + C_3 \Rightarrow C_3 = 0$$

$$0 = y = -\frac{g(0)^2}{2} + V(0) \sin \theta + C_4$$

$$\Rightarrow C_4 = 0$$

$$\therefore x = Vt \cos \theta, \quad y = -\frac{gt^2}{2} + Vt \sin \theta$$

Now want horizontal range to be at least c metres.Range Max when $y=0$.

$$\therefore 0 = -\frac{gt^2}{2} + Vt \sin \theta$$

$$0 = t \left(-\frac{gt}{2} + V \sin \theta \right)$$

$$\therefore t=0 \text{ or } \frac{gt}{2} = V \sin \theta$$

$$\therefore t = \frac{2V \sin \theta}{g}$$

Q16

a) cont'd

$$\therefore x = Vt \cos \theta$$

$$= V \left(\frac{2V \sin \theta}{g} \right) \cos \theta$$

$$= \frac{V^2 \sin 2\theta}{g}$$

$$\text{Now } \frac{V^2 \sin 2\theta}{g} > c$$

Since Max $\sin 2\theta = 1$ then

$$\frac{V^2}{g} > c$$

(4)

$$\therefore V^2 > cg$$

Since $V, c, g > 0$ then

$$V > \sqrt{cg} \quad \text{QED!}$$

b) i) RTP:

$$1 + r + r^2 + \dots + r^{n-1} = \frac{1-r^n}{1-r}, \quad r \neq 1$$

$$1 + r + r^2 + \dots + r^{n-1} = \frac{a(1-r^n)}{1-r}$$

Since G.P. now $a=1$ so

$$1 + r + r^2 + \dots + r^{n-1} = \frac{1(1-r^n)}{1-r}$$

$$\left[\begin{array}{l} \text{G.P. since} \\ \frac{T_3}{T_2} = \frac{r^2}{r}, \frac{T_2}{T_1} = \frac{r}{1} \\ \quad \quad \quad = r \quad \quad \quad = r \end{array} \right] = \frac{1-r^n}{1-r} \quad \text{QED!} \quad (r \neq 1)$$

(1)

$$\text{ii) } \lim_{n \rightarrow \infty} 1 + r + r^2 + \dots + r^{n-1} = \lim_{n \rightarrow \infty} \frac{1-r^n}{1-r}$$

$$= \frac{1-0}{1-r}$$

for $0 < r < 1$

$$= \frac{1}{1-r}$$

(1)

16b) cont'd

iii) For $0 < x < 1$

$$\text{RTP: } x + \frac{x^2}{2} + \frac{x^3}{3} + \dots + \frac{x^n}{n} + \dots = -\ln(1-x).$$

Proof: From (ii)

$$1 + r + r^2 + \dots + \dots = \frac{1}{1-r} \text{ for } 0 < r < 1$$

$$\therefore \int_0^x (1 + r + r^2 + \dots + \dots) dr = \int_0^x \frac{1}{1-r} dr$$

$$\therefore \left[r + \frac{r^2}{2} + \frac{r^3}{3} + \dots + \frac{r^n}{n} + \dots \right]_0^x = \left[-\ln|1-r| \right]_0^x$$

$$\therefore x + \frac{x^2}{2} + \frac{x^3}{3} + \dots + \frac{x^n}{n} + \dots - (0 + 0 + 0 + \dots)$$

$$= -[\ln|1-x| - \ln|1-0|]$$

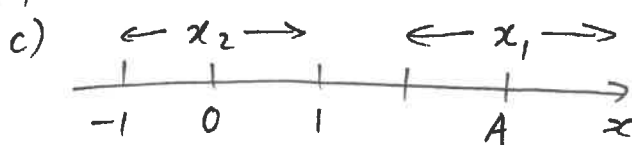
$$\therefore x + \frac{x^2}{2} + \frac{x^3}{3} + \dots + \frac{x^n}{n} + \dots = -\ln|1-x| + 0$$

QED !!!

$$= -\ln|1-x|.$$

(2)

Q16 cont'd



$$x_1 = A + \sin nt$$

$$\ddot{x}_2 = -n^2 x_2$$

$$t=0, x_1 = A, x_2 = 1.$$

i) RTP: $\dot{x}_1^2 = n^2(1 - (x_1 - A)^2)$

Proof: $x_1 = A + \sin nt$

$$\dot{x}_1 = n \cos nt$$

$$\therefore \dot{x}_1^2 = n^2 \cos^2 nt$$

$$x_1 - A = \sin nt$$

$$\therefore (x_1 - A)^2 = \sin^2 nt$$

Now $\cos^2 nt + \sin^2 nt = 1$

$$\therefore \dot{x}_1^2 = n^2 \cos^2 nt$$

$$\therefore \dot{x}_1^2 = n^2(1 - \sin^2 nt)$$

$$= n^2(1 - (x_1 - A)^2)$$

QED!!!

ii) RTP: $\dot{x}_2^2 = n^2(1 - x_2^2)$

Proof: $\ddot{x}_2 = -n^2 x_2$

$$\therefore \frac{d}{dx} \left(\frac{1}{2} v_2^2 \right) = -n^2 x_2 \quad (2)$$

$$\frac{1}{2} v_2^2 = \int -n^2 x_2 dx_2$$

$$\frac{1}{2} v_2^2 = -\frac{n^2 x_2^2}{2} + C$$

When $t=0, x_2 = 1, v_2 = 0$:

$$0 = -\frac{n^2(1)^2}{2} + C$$

$$\therefore C = \frac{n^2}{2}$$

Q16 c) ii) cont'd:

$$\therefore \frac{1}{2} v_2^2 = -\frac{n^2 x_2^2}{2} + \frac{n^2}{2}$$

$$\therefore v_2^2 = -n^2 x_2^2 + n^2$$

$$\therefore v_2^2 = n^2(1 - x_2^2)$$

$$\therefore \dot{x}_2^2 = n^2(1 - x_2^2) \quad \text{QED!!}$$

iii) Since $\ddot{x}_2 = -n^2 x_2$ we know the centre of motion is 0. Also, when $t=0, v_2 = 0$ When $x_2 = 1 \therefore P_2$ is oscillating around 0 with amplitude 1 starting at 1 $\therefore$ convenient equation is $x_2 = \cos nt$.

Since x_1 is oscillating around A where $A > 0$ then minimum distance is given by $x_1 - x_2$ minimised.

$$\therefore x_1 - x_2 = A + \sin nt - \cos nt$$

$$\frac{d}{dt}(A + \sin nt - \cos nt) = 0$$

$$n \cos nt + n \sin nt = 0 \quad (3)$$

$$\sin nt = -\cos nt$$

$$\tan nt = -1 \quad t \geq 0$$

$$nt = \frac{3\pi}{4}, \frac{7\pi}{4}, \frac{11\pi}{4}, \frac{15\pi}{4}, \dots$$

$$t = \frac{3\pi}{4n}, \frac{7\pi}{4n}, \frac{11\pi}{4n}, \frac{15\pi}{4n}, \dots$$

We can see from sketch that minimum is when

$$t = \frac{7\pi}{4n}$$

PTD

Q16 cont'd

c) iii) cont'd

 $\therefore$ Min distance

$$= A + \sin\left(\frac{7\pi}{4}\right) - \cos\left(\frac{7\pi}{4}\right)$$

$$= A + \frac{-1}{\sqrt{2}} - \frac{1}{\sqrt{2}}$$

$$= A - \frac{2}{\sqrt{2}}$$

$$= A - \sqrt{2} \text{ assuming } A > \sqrt{2}!$$

